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Microstructure and Mechanical Properties of Dissimilar Metal Welded Joints for Nuclear Applications Using Laser Welding with Dissimilar Filler Wire Addition

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Abstract

Laser welding with wire filler addition has been a potential manufacture technique for the nuclear pressure vessels manufacturing. However, laser welding by using only single nickel-based alloy wire or austenitic stainless steel wire still faces some challenges, such as non-fusion, cracks, low mechanical properties and C element migration. Thus, we assess the possibility of filling 52 M and 308 l dissimilar wires for a narrow gap laser welding of SA508 and 316 l dissimilar metals in nuclear pressure vessels fabrication in this paper. The results showed that with increasing the ratio of 308 l filling wire from 25 to 75%, the fusion line boundary of 52 M/Weld metal gradually became obvious and the size of the cellular and dendritic grains increased slightly in the weld metal. The results also indicated that the average yield strength (YS) of the welded joints obtained with different ratio of the filling double wires was about 322 MPa, which was between that of the 316 l stainless steel and the SA508 low alloy ferritic steel base metals. Additionally, the average ultimate tensile strength (UTS) of the joints obtained with different ratio of the double filling wires was about 571 MPa, which was lower than that of the 316 l and the SA508 base metal. The maximum UTS of the welded joint was achieved when the 52 M/308L filling wires ratio was 3:1. The maximum impact fracture absorbed energy of the welded joint was obtained when the 52 M/308L filling wires ratio was 1:3. When the ratio of the 308 l filling wire increased, the impact fracture absorbed energy gradually increased.

Keywords Dissimilar metals · Narrow gap laser welding · Microstructure · Tensile strength · Impact toughness

1 Introduction

Reactor pressure vessel is a key part in a nuclear power plant [1]. There are many dissimilar metal welded joints in the connecting of reactor pressure vessels and pipes [2]. At present, low alloy ferritic steel SA508 is a material universally used as reactor pressure vessel, while the material of pipe is usually austenitic stainless steel 316 l. In a primary circuit and safety end of a nuclear reactor pressure vessel (Fig. 1), welding problems such as carbon migration, high residual strain [3, 4] and cracking [5–7] are prone to occur in the welded joints due to the wide differences in the physical properties and the chemical composition between the SA508 and the 316 l [8, 9].

Austenitic stainless steel welding wire as well as nickel based alloy welding wire are the two main welding materials for welding of dissimilar metals in nuclear power plants.

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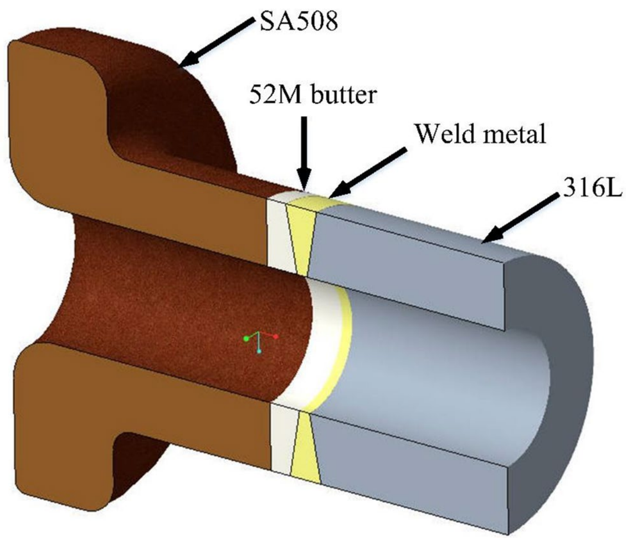


Fig. 1 Dissimilar metal welded joints in reactor pressure vessel

However, dissimilar metal welded joints for nuclear applications could not simultaneously take into account all the weld quality requirement by only using single welding wire of a nickel-based alloy welding wire or an austenitic stainless steel welding wire.

Initially, stainless steel welding wire was the common wire for welding dissimilar metals welded joint for nuclear power manufacturing. Ming et al. [10] reported that 309 l and 308 l wire were adopted as filling welding wire for the SA508 and the 316 l stainless steel dissimilar metals welding process. The plasticity and strength of the joints were reduced because of the discrepancy in physical and chemical properties between 308 l/309L stainless steel welding wire and low alloy ferritic steel SA508 base metal. Meanwhile, Xiong et al. [11] indicated that carbon migration was occurred in the welded joint of SA508 and 316 dissimilar metals. Guo et al. [12] presented that near the fusion line, the coarse grained zone was composed of coarse martensite and tempered martensite, while the fine grained zone consisted of small martensite and tempered martensite in the laser welding of SA508. Additionally, the microhardness of the welded joint was twice higher than that of the base metals. Suranjit et al. [13] proposed that when the butter was prepared on one side of the low alloy steel, large internal stress would appear at the side of the low alloy steel. The stress removal treatment would promote carbon migration in their research. Furthermore, Wang et al. [14–16] reported that obvious epitaxial growth and competitive growth were obtained in the interface of weld metal and 316 l base metal in the welding of low alloy ferrite steel and stainless steel by using an austenite stainless steel welding wire. Additionally, the interface was more sensitive to stress corrosion cracking (SCC).

In order to eliminate the problems (such as low strength, carbon migration and sensitive to the SCC) of using stainless steel welding wire as filler material in the welding of dissimilar metals, Ranjbar et al. [17] introduced a nickel-based alloy filling wire and 309 l, respectively, for joining nickel based alloy and low alloy steel dissimilar metals. Their research found that compared with stainless steel wire, nickel based alloy filling wire had an excellent ability to resist the migration of carbon element. Later, Santosh et al. [18] compared 309 l and FM82 as the filling materials and reported that when FM82 was used as a filling material, the phenomenon of carbon element migration near the welding fusion line interface was significantly inhibited. Those experimental results were consistent with the investigation by Ghosh et al. [19] and Orr et al. [20]. However, Platt et al. [21] presented that the SCC sensitivity of FM82 weld was relatively high due to the precipitation of carbides such as $Cr_{23}C_6$ and Cr_7C_3 along the grain boundary. Ming et al. [22] found that when FM52 was used as a filler, there was a dilution zone near the fusion line close to SA508. Kim et al. [23] also found that when a nickel-based alloy 82/182 was used as a filling material, there was no obvious migration of carbon on the side of the low alloy steel, but there were different microstructures in the heat affected zone of SA508, and significant gradient changes in mechanical properties near the heat affected zone. However, Ahonen et al. [24] and Oberson et al. [25] demonstrated that welding with FM52 filler wire for dissimilar metal welded joints was prone to high temperature ductility dip cracking. Then, Zhu et al. [26] demonstrated that large residual strain was obtained at the 52 M/316L interface, and that the rate of crack extension in the 316 l heat affected zone was positively correlated with temperature and dissolved oxygen content. Lu et al. [27] also reported that the residual stress at the interface of 316 l was the largest and that increasing the distance to the fusion line, the residual stress diminished.

In conclusion, for the welding of low alloy steel and stainless steel dissimilar metals, the welding wire was mainly nickel-based alloy wire or stainless steel wire. Welding with stainless steel wire was easy to cause carbon migration at the fusion line interface. Meanwhile, the interface was more sensitive to stress corrosion cracking (SCC) of the welded joint decreased with filling stainless steel wire. However, welding with a nickel base alloy wire, high residual stress appeared at the welded joint interface. Additionally, the nickel base alloy molten metal presented large viscosity, poor fluidity and wetting, which would cause lack of fusion in the welding process.

Thus, in this paper we assess the possibility of using simultaneously both 52 M and 308 l wires for the narrow gap laser welding (NGLW) of SA508 and 316 l dissimilar metals for the manufacture of critical nuclear pressure vessels.

Table 1 Chemical composition of base metals and welding wires (wt%)

	C	Cr	Ni	Mn	Si	Cu	S	*P*	Mo	V	Fe
SA508	0.16	0.12	0.75	1.47	0.21	0.02	0.003	0.004	0.50	0.009	Bal.
316 l	0.028	18.01	12.1	1.77	0.06	0.17	0.006	0.015	2.43	/	Bal.
52 M	0.023	30.16	Bal.	0.01	0.04	0.01	0.001	0.009	0.01	/	8.54
308 l	0.027	20.49	10.03	1.70	0.03	0.08	0.002	0.039	0.08	/	Bal.

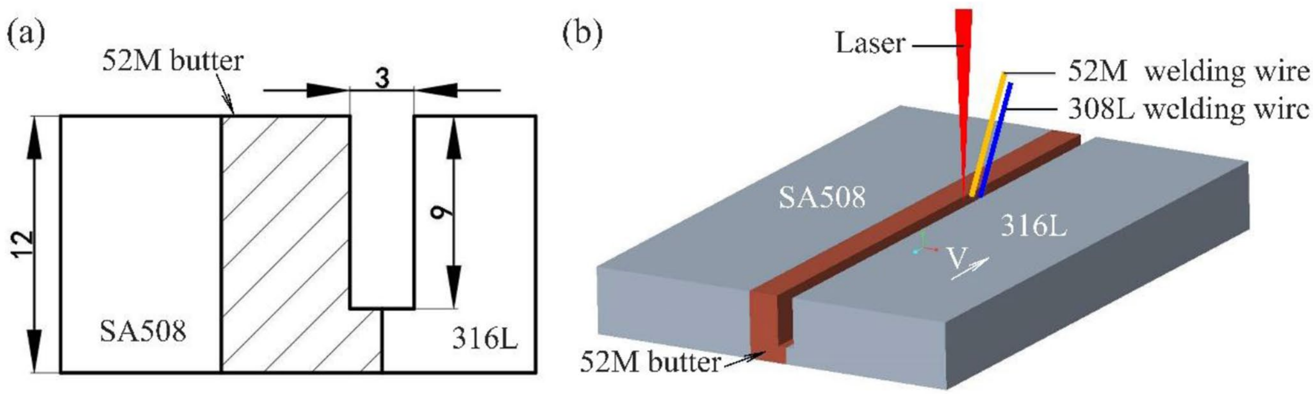


Fig. 2 The dissimilar metals welded joint with filling double wires: **(a)** groove and **(b)** welding process diagram

Table 2 Parameters of the laser welding process

Pass number	*P* (kW)	D (mm)	V_{wire} (mm/min)	$V_{welding}$ (mm/min)
1	5.2	0	/	1800
2	4.8	60	200	480
3	4.8	60	300	480
4	4.8	60	400	480
5	4.8	60	500	480

The effect of the microstructure and crystallographic characteristics on the tensile and impact toughness properties of the welded joint was established in details.

2 Experimental Procedure

Low alloy ferritic steel SA508 and stainless steel 316 l were used as dissimilar base metals. The thickness of the base metals were 12 mm. Additionally, FM52M (52 M) and 308 l were used as the filling welding wires. The diameter of the wires were 1.0 mm. The chemical composition of the base metals and the welding filling wires are shown in Table 1. The five ratio of 52 M and 308 l wires were: 75%:25%, 60%:40%, 50%:50%, 40%:60% and 25%:75%. The five ratio were obtained by changing the feed velocity of the two wires. At the ratio of 50%:50% of the 52 M and 308 l wires, both of the feed velocity of the wires were 100 mm/min. A 99.99% argon gas was used as a shielding gas. The flow rate of the shielding gas was 20 l/min.

A butt joint configuration is designed and the groove is I, as shown in Fig. 2. The 52 M welding wire is near the 52 M butter side, while the 308 l welding wire is on the 316 l base metal side. The root face of the joint is 3 mm [28–30]. A 52 M butter on the welding side of the SA508 base metal was prepared by using a laser cladding. The thickness of the 52 M butter was 7.5 mm. A five passes welding sequence was developed. The backing welding was laser self-fusion welding. The other four filling welding was laser welding with filling double wires.

An IPG laser systems and two Fronius wire feeders were used in the welding process. The maximum power of the laser was 6 kW. The wavelength of the laser was 1.06 µm. The diameter of the focus laser spot was 0.6 mm. The forward tilt angle of the defocus laser beam (α) is about 6°. The spacing between the laser beam and the wire is -1 mm. The laser power (P), defocus length (D), wire feeding velocity (V_{wire}), welding velocity ($V_{welding}$) of the narrow gap multi-pass laser welding are shown in Table 2.

An OLYMPUS BX53M optical microscope was used to analyze the metallurgical specimens. A HITACHI SU8230 model scanning electron microscopy (SEM) was used to analyze the microstructure and the fracture morphology feature. The scanning step of the electron backscattered diffraction (EBSD) tests was 0.8 µm.

For the tensile strength analysis, the thickness of the tensile specimen is 12 mm. The tensile testing machine was MTS C45.305 with loading velocity of 1 mm/min at temperature of 25 ℃. The fracture surfaces is observed [31]. Additionally, an Instron-9250 impact testing machine was used for the Charpy V-notch impact testing specimens at temperature of 25 ℃. The schematic diagram of the tensile specimens and impact specimens is describe in Fig. 3.

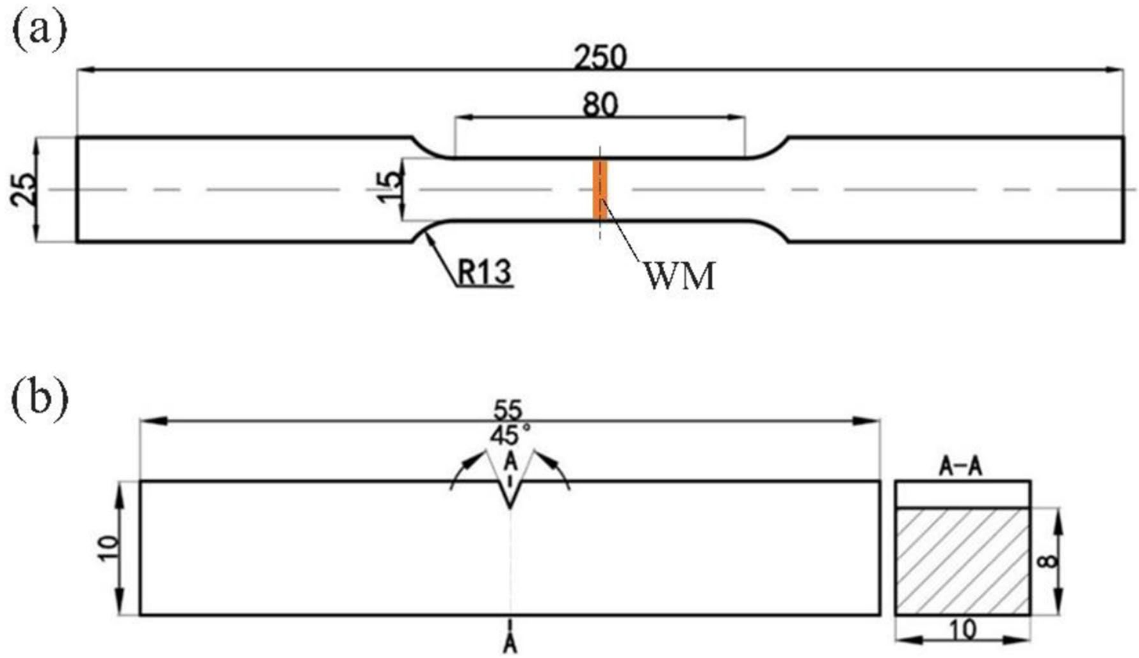
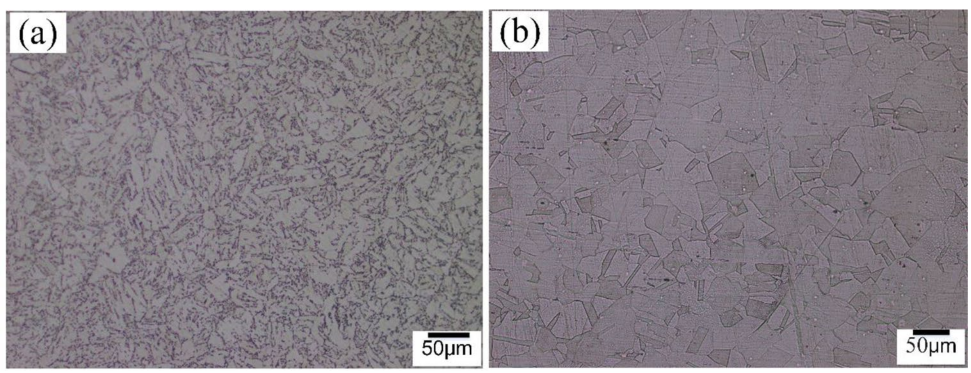


Fig. 3 Schematic of: **(a)** tensile specimen and **(b)** impact specimen

Fig. 4 Microstructure: **(a)** SA508 and **(b)** 316 l



3 Results and Discussion

3.1 Microstructure

In our previous research [28], welded joint without cracks, porosities, lack of fusion defects to sidewalls and between layers could be obtained by a narrow gap. The microstructure of the SA508/316L dissimilar base metals are described in Fig. 4. The SA508 base metal constitutes of ferrite and pearlite which located at the grain boundaries, while the 316 l base metal is composed of austenite. Some twin crystal are also found in the 316 l base metal.

Figure 5 presents the fusion line microstructure of 52 M and WM. It can be seen that both the 52 M butter and the WM are austenitic structure with cell crystal and dendritic crystal because the 52 M filling wire and 308 l filling wire are both austenitic structure. The grow direction of the dendritic crystal is perpendicular to the welding direction with maximum temperature gradient during the solidification process.

Figure 5 also presents that with the increasing ratio of stainless steel 308 l welding wire from 25 to 75%, the 52 M and WM fusion line boundary gradually become obvious.

Figure 6 presents the microstructure of the WM. It can be seen that with the increasing ratio of stainless steel 308 l welding wire (from 25 to 75%) in the double wires filler addition, the size of the cellular and dendritic grains increases slightly in the WM.

3.2 Tensile Strength

Figure 7 describes the stress strain relation curves of the tensile specimens of the base metal and welded joints. The figure shows that the average yield strength (YS) of the welded joints with different ratio of 52 M and 308 l filling wires is about 322 MPa, which is between that of 316 l and SA508

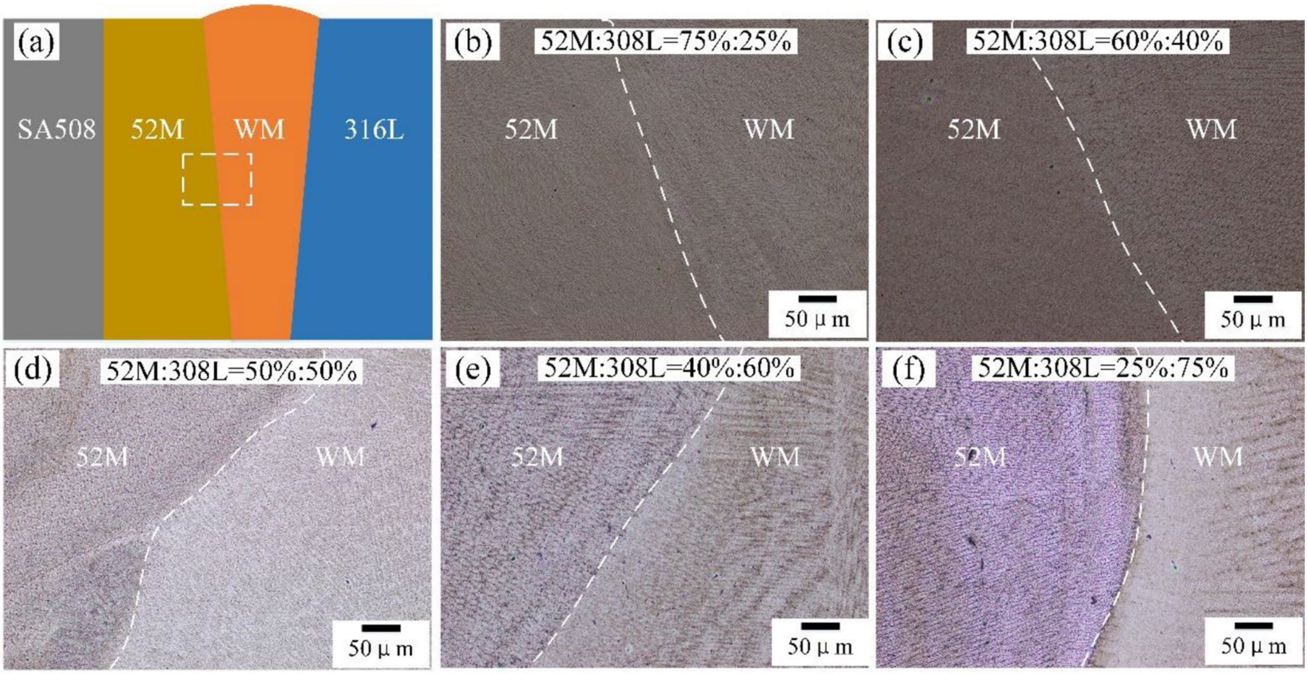


Fig. 5 Microstructures of the 52 M/WM interface with different 52 M/308L wires ratio: **(a)** diagram, **(b)** 75%:25%, **(c)** 60%:40%, **(d)** 50%:50%, **(e)** 40%:60% and **(f)** 25%:75%

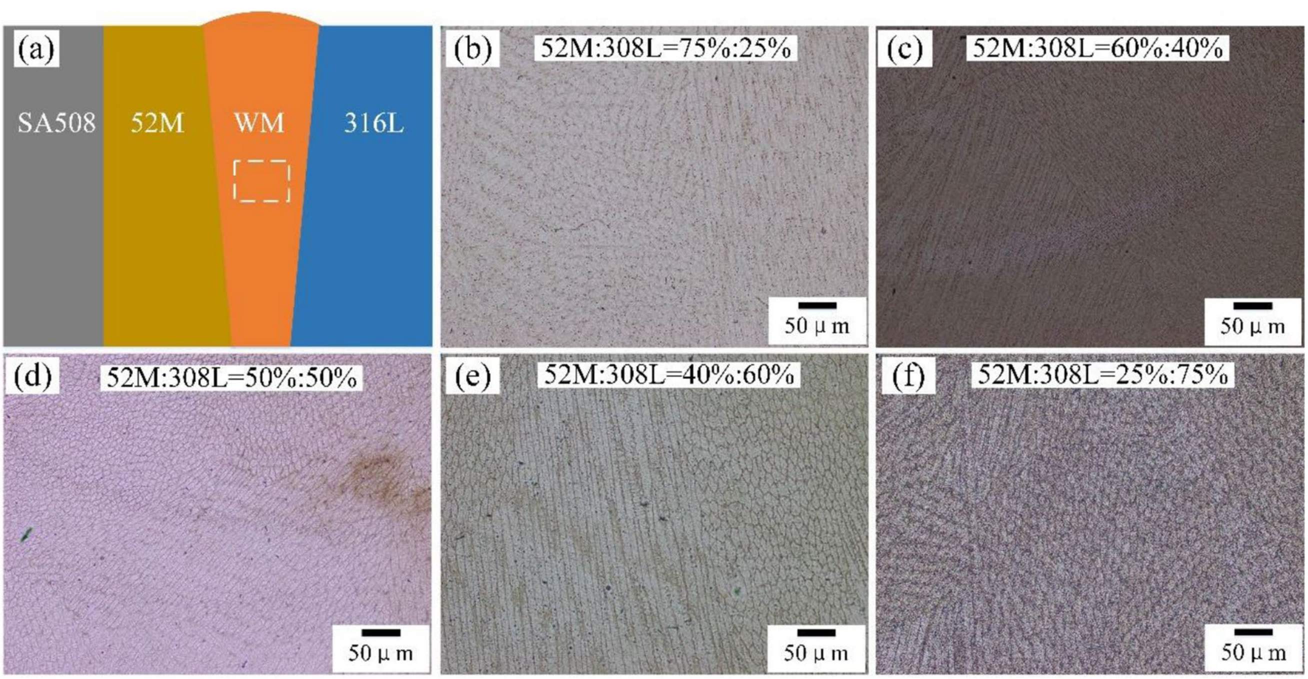


Fig. 6 Microstructures of the WM with different 52 M/308L wires ratio: **(a)** diagram, **(b)** 75%:25%, **(c)** 60%:40%, **(d)** 50%:50%, **(e)** 40%:60% and **(f)** 25%:75%

base metal, with value of 316 and 456 MPa, respectively. Additionally, the average ultimate tensile strength (UTS) of the welded joints with different ratio of 52 M and 308 l filling wires is about 571 MPa, which is lower than that of 316 l base metal and SA508 base metal, with value of 606 and 574 MPa, respectively. It should be noted that the maximum UTS of the welded joint was achieved when the filling wires ratio was 75% 52 M:25% 308 l, which meant that increasing the ratio of the 52 M filling wire could improve the tensile strength of the welded joint.

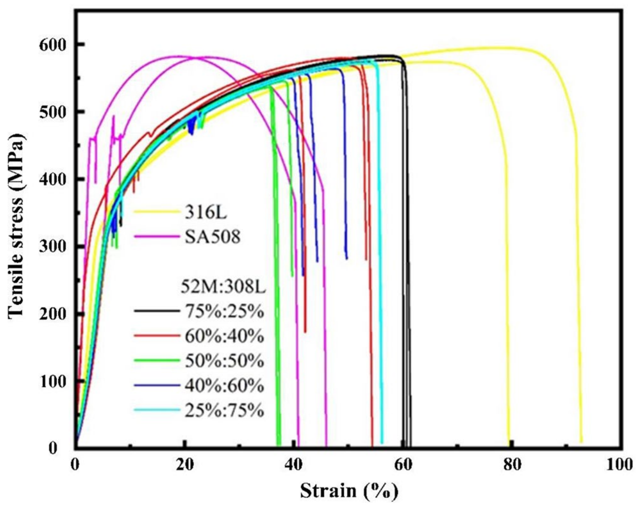


Fig. 7 Stress-strain relationship the SA508 base metal, 316 l base metal and welded joints with different ratio of filling wires

As shown in Fig. 8, for all the tensile specimens of the welded joints, the fracture locations are at the WM side near the WM/316L interface, although both WM and 316 l are austenitic structure. To understand the fracture locations of the welded joint, the crystallographic features of WM/316L interface with ratio of filling wires of 75% 52 M: 25% 308 l, 50% 52 M: 50% 308 l and 25% 52 M: 75% 308 l were examined by EBSD. The inverse pole figure (IPF) in Fig. 9 shows that the average effective grain sizes of all the three WM are larger than those of the 316 l. Small grain size could promote the YS and UTS of the tensile specimens according to the Hall-Petch relationship. Thus, the fracture locations of the welded joint occurred at the WM side near the WM/316L interface due to the large grain size of the WM. Additionally, as the coefficient of thermal expansion of 316 l is higher than that of the 52 M, the residual stress at the interface of WM/316L is larger than that of the WM/52 M. Thus, fracture is more likely to occur at the interface of WM/316L but not the WM/52 M, because under

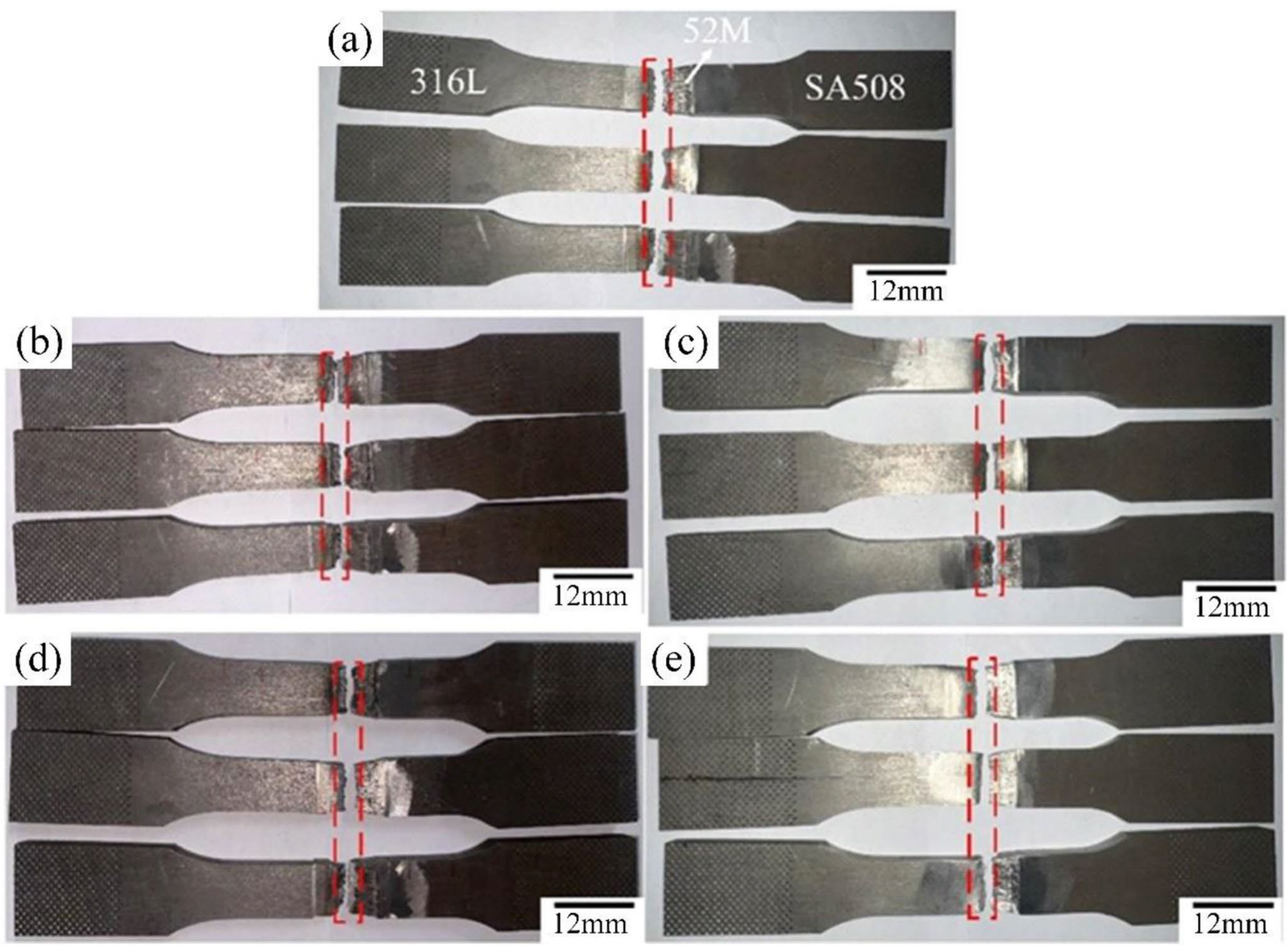


Fig. 8 Fracture location of the tensile specimens of the welded joint with different 52 M/308L wires ratio: **(a)** 75%:25%, **(b)** 60%:40%, **(c)** 50%:50%, **(d)** 40%:60% and **(e)** 25%:75%

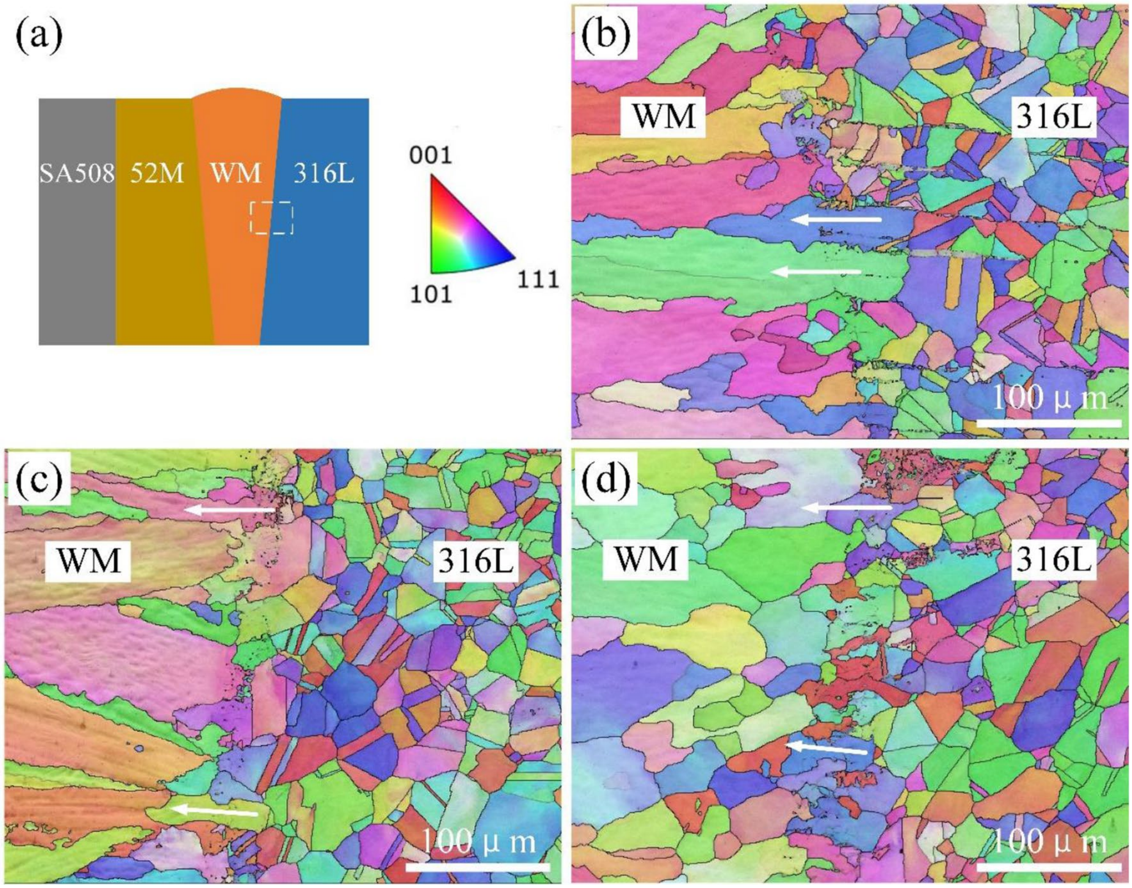


Fig. 9 IPF of the grain boundary distribution with high angle of misorientation (> 15°) of the WM/316L interface of the welded joint with different 52 M/308L wires ratio: **(a)** diagram, **(b)** 75%:25%, **(c)** 50%:50% and **(d)** 25%:75%

the effect of residual stress, crack initiation and propagation are more likely to occur at the interface of WM/316L.

Additionally, Fig. 9 also shows that prominent epitaxial growth occurs along the WM and 316 l fusion line and that the grain growth orientation (white arrow marks) of the WM is consistent with that of the 316 l base metal due to all the microstructure of filling wire 52 M, 308 l and 316 l base metal are austenitic structure. The 316 l grain is almost completely wetted with the liquid metal in the weld pool because of the same crystal structure.

Figure 10 presents the fracture morphology of the tensile specimen. The fracture morphologies show typical dimple-like structures. Thus, the ductile fracture is the main fracture mode of all the tensile specimens.

3.3 Impact Toughness

The absorbed energies of impact fracture of the 52 M butter, 52 M/WM interface, WM, WM/316L interface with different ratio of 52 M and 308 l filling wires and 316 l base metal are shown as curves in Fig. 11. It can be observed that all impact absorbed energy values of the welded joints are the lower than the corresponding values of the 52 M butter, 52 M/WM interface, WM/316L interface and 316 l base metal.

The Fig. 11 also indicates that when the filling wires ratio is 25% 52 M:75% 308 l, the maximum impact fracture absorbed energy of the welded joint is obtained. Additionally, when the ratio of the 308 l increases, the impact fracture absorbed energy gradually increases, which indicates that 308 l wire could improve the impact fracture absorbed energy of the welded joint.

4 Conclusions

An innovative approach of applying 52 M and 308 l dissimilar filling wires for SA508/316L dissimilar metals in the NGLW for critical nuclear pressure vessels manufacturing was developed. The following conclusions were obtained:

(1) Epitaxial growth occurred along the WM and 316 l fusion line and the grain growth orientation of the WM was consistent with the grain orientation of 316 l base

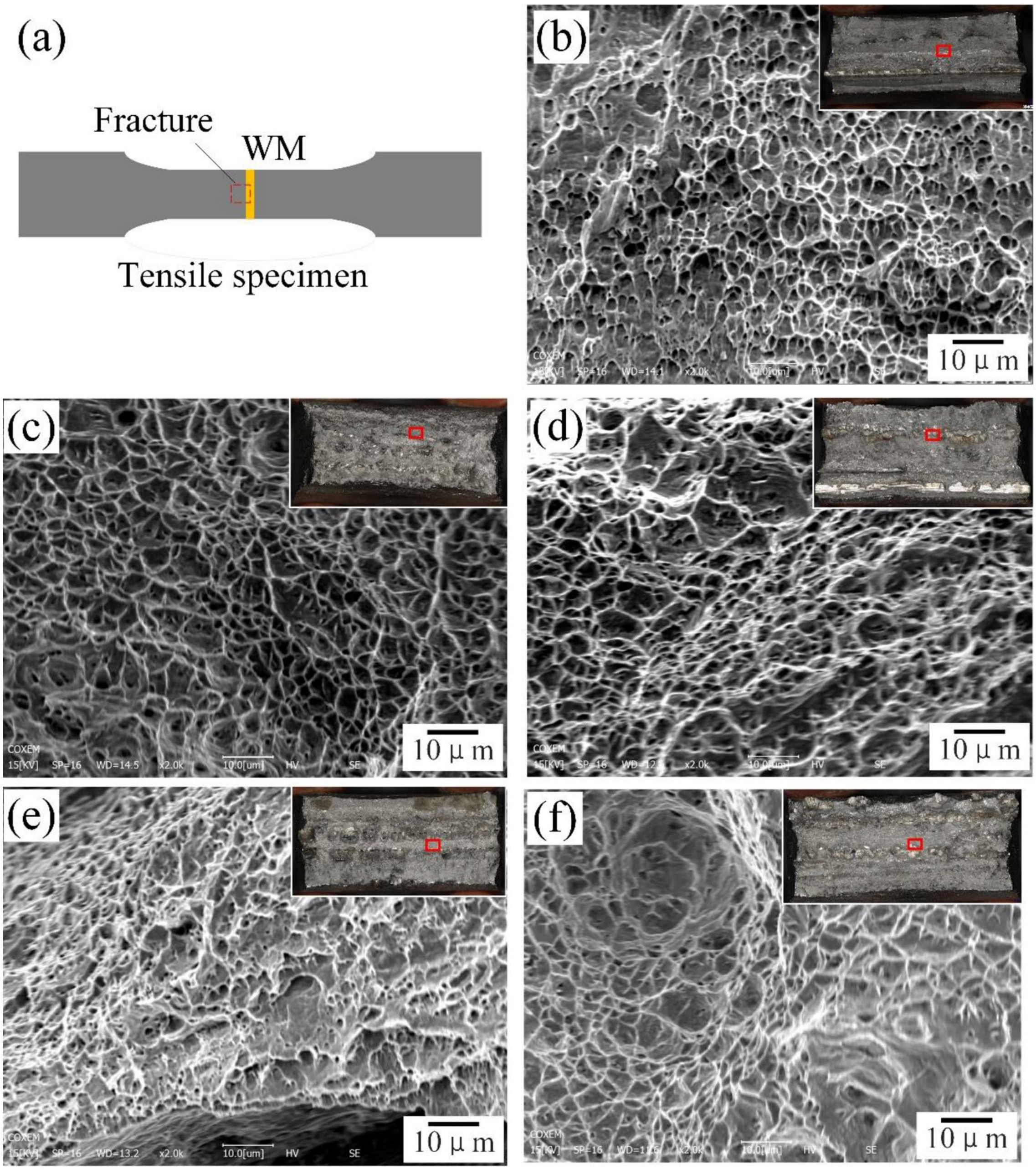


Fig. 10 Fracture morphologies of the tensile specimens of welded joints with different 52 M/308L wires ratio: **(a)** schematic diagram of fracture location, **(b)** 75%:25%, **(c)** 60%:40%, **(d)** 50%:50%, **(e)** 40%:60% and **(f)** 25%:75%

metal, because that all the microstructure of nickel-based alloy welding wire 52 M, stainless steel welding wire 308 l and base metal 316 l were austenitic structure. The 316 l grain was wetted to the liquid welded metal, and the liquid welded metal atoms in the melt pool would maintain the original grain orientation due to the same crystal structure,

(2) The average YS of the welded joints with different ratio filling wires was about 322 MPa, which was between that of 316 l base metal and SA508 base metal, with value of 316 and 456 MPa, respectively. Additionally, the average UTS of the welded joints with different ratio of filling wires was about 571 MPa, which was lower than that of 316 l base metal and SA508 base metal, with value of 606 and 574 MPa, respectively. The maximum UTS of the welded joint was achieved when the filling wires ratio was 75% 52 M:25% 308 l.

(3) All the absorbed energy of the impact of the welded joints were lower than those of the 52 M butter, 52 M/WM interface, WM/316L interface and 316 l base

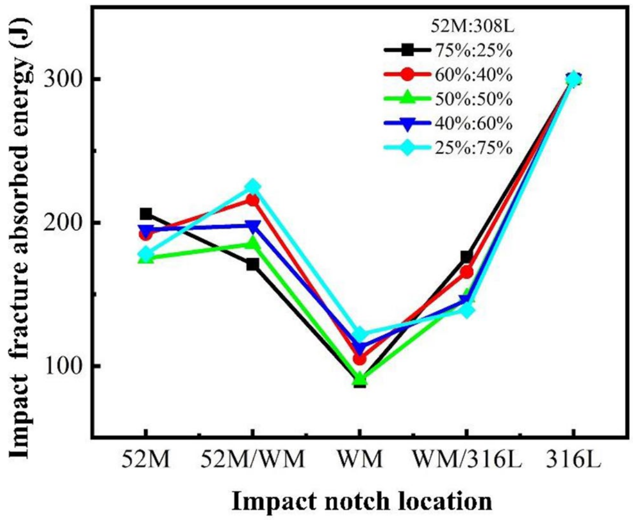


Fig. 11 Impact toughness of the 52 M butter, 52 M/WM interface, WM, WM/316L interface and 316 l base metal

metal. The maximum impact fracture absorbed energy of the welded joint was obtained when the filling wires ratio was 25% 52 M:75% 308 l. Additionally, when the 308 l filling wire increased, the impact fracture absorbed energy gradually increased, which indicated that the 308 l wire could improve the impact fracture absorbed energy of the welded joint.

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Declarations

Declaration of Generative AI No artificial intelligence was used in the manuscript.

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. I. Medouni, A. Portavoce, P. Maugis, M. Descoins, M. Yescas, F. Roch, P. Joly, K. Hoummada, Thermal ageing effect on solute segregation and precipitation in the heat-affected-zone of dissimilar metal welds for nuclear power plants. Mater. Des. **232**, 112158 (2023). https://doi.org/10.1016/j.matdes.2023.112158
2. Y. Ren, R. Skilton, A review of pipe cutting, welding, and NDE technologies for use in fusion devices. Fusion Eng. Des. **202**, 114396 (2024). https://doi.org/10.1016/j.fusengdes.2024.114396
3. Z. Gao, L. Zhao, Y. Han, Investigation on residual stress in SA508/Inconel metal/CF8A dissimilar welded joint for nuclear steam, generator safe end using different processes. Nucl. Technol. **210**, 471–485 (2023). https://doi.org/10.1080/00295450.2023.2229602
4. W. Jiang, W. Xie, X. Qi, Y. Deng, Y. Wan, X. Xie, Residual stress and microstructure control in welding of SA508 low alloy steel. Int. J. Press. Vessels Pip. **207**, 105090 (2024). https://doi.org/10.1016/j.ijpvp.2023.105090
5. S. Nuthalapati, K.E. Kee, S.R. Pedapati, K. Jumbri, A review of chloride induced stress corrosion cracking characterization in austenitic stainless steels using acoustic emission technique. Nucl. Eng. Technol. **56**, 688–706 (2024). https://doi.org/10.1016/j.net.2023.11.005
6. M. Koshiishi, D. Akazawa, Y. Miura, K. Kako, Prediction of stress corrosion crack growth rate of cold worked stainless steels in BWR conditions by the slip oxidation model. Nucl. Eng. Des. **421**, 113060 (2024). https://doi.org/10.1016/j.nucengdes.2024.113060
7. A.A. Amiri, I.U. Toor, A. Shams, Understanding stress corrosion cracking (SCC), affecting variables and prevention strategies in nuclear power plants-a review. Arab. J. Sci. Eng. **50**, 2957–2990 (2025). https://doi.org/10.1007/s13369-024-09241-w
8. J. Feng, W. Guo, J. Francis, N. Irvine, L. Li, Narrow gap laser welding for potential nuclear pressure vessel manufacture. J. Laser Appl. **28**, 022421 (2016). https://doi.org/10.2351/1.4943905
9. L.T. Zhang, J.Q. Wang, Effect of dissolved oxygen content on stress corrosion cracking of a cold worked 316L stainless steel in simulated pressurized water reactor primary water environment. J. Nucl. Mater. **446**, 15–26 (2014). https://doi.org/10.1016/j.jnucmat.2013.11.027
10. H. Ming, Z. Zhang, J. Wang, E.H. Han, W. Ke, Microstructural characterization of an SA508–309L/308L–316L domestic dissimilar metal welded safe-end joint. Mater. Charact. **97**, 101–115 (2014). https://doi.org/10.1016/j.matchar.2014.08.023
11. Q. Xiong, H. Li, Z. Lu, J. Chen, Q. Xiao, J. Ma, X. Ru, Characterization of microstructure of A508III/309L/308L weld and oxide films formed in deaerated high-temperature water. J. Nucl. Mater. **498**, 227–240 (2018). https://doi.org/10.1016/j.jnucmat.2017.10.030
12. W. Guo, S. Dong, W. Guo, J.A. Francis, L. Li, Microstructure and mechanical characteristics of a laser welded joint in SA508 nuclear pressure vessel steel. Mater. Sci. Eng. A **625**, 65–80 (2015). https://doi.org/10.1016/j.msea.2014.11.056
13. S. Kumar, R.P. Kale, P.K. Singh, M. Ghosh, J. Chattopadhyay, Fracture toughness behavior of dissimilar metal (SA508 Gr.3 class 1 and SA312 type 304LN) weld joint: with and without stress relieving treatment, fatigue fract. Eng. Mater. Struct. **44**, 2462–2474 (2021). https://doi.org/10.1111/ffe.13524
14. Z. Wang, J. Xu, T. Shoji, Y. Takeda, H. Yuya, M. Ooyama, Microstructure and pitting behavior of the dissimilar metal weld of 309L cladding and low alloy steel A533B. J. Nucl. Mater. **508**, 1–11 (2018). https://doi.org/10.1016/j.jnucmat.2018.05.011
15. A. Alsaran, K. Celik, Mechanical and structural properties of similar and dissimilar steel joints. Mater. Charact. **43**, 311–318 (1999). https://doi.org/10.1016/S1044-5803(99)00045-5
16. G.F. Li, E.A. Charles, J. Congleton, Effect of post weld heat treatment on stress corrosion cracking of a low alloy steel to stainless steel transition weld. Corros. Sci. **43**, 1963–1983 (2001). https://doi.org/10.1016/S0010-938X(00)00182-7
17. K. Ranjbar, R. Dehmolaei, M. Amra, I. Keivanrad, Microstructure and properties of a dissimilar weld between alloy 617 and

A387 steel using different filler metals. Weld. World **62**, 1121–1136 (2018). https://doi.org/10.1007/s40194-018-0610-x
18. R. Santosh, G. Das, S. Kumar, P.K. Singh, M. Ghosh, Experimental and computational investigation of structural integrity of dissimilar metal weld between ferritic and austenitic steel. Metall. Mater. Trans. A **49**, 2099–2112 (2018). https://doi.org/10.1007/s11661-018-4554-y
19. A. Ghosh, M. Sahu, P.K. Singh, S. Kumar, M. Ghosh, S.P. Sagar, Assessment of mechanical properties for dissimilar metal welds: A nondestructive approach. J. Mater. Eng. Perform. **28**, 900–907 (2019). https://doi.org/10.1007/s11665-019-3867-3
20. M.R. Orr, J.C. Lippold, F. Argentine, Evaluation of solidification cracking susceptibility in ERNiCr-3 (Filler metal 82) weld metal using the cast pin tear test. Weld. World **61**, 935–944 (2017). https://doi.org/10.1007/s40194-017-0473-6
21. P. Platt, J. Sayers, D.A. Horner et al., Hydrogen-induced brittle fracture in nickel based alloy 82 weld metal. Corros. Sci. **153**, 118–126 (2019)
22. H. Ming, Z. Zhang, J. Wang, E.H. Han, Microstructure of a domestically fabricated dissimilar metal weld joint (SA508-52 M-309L-CF8A) in nuclear power plant. Mater. Charact. **148**, 100–115 (2019). https://doi.org/10.1016/j.matchar.2018.12.011
23. J.W. Kim, K. Lee, J.S. Kim, T.S. Byun, Local mechanical properties of alloy 82/182 dissimilar weld joint between SA508 Gr.1a and F316 SS at RT and 320°C. J. Nucl. Mater. **384**, 212–221 (2009). https://doi.org/10.1016/j.jnucmat.2008.11.019
24. M. Ahonen, S. Lindqvist, T. Sarikka et al., Effect of thermal aging on fracture mechanical properties and crack propagation behavior of alloy 52 narrow-gap dissimilar metal weld. in *Proceedings of the 18th International Conference on Environmental Degradation of Materials in Nuclear Power Systems-Water Reactors*, ed. by J.H. Jackson, D. Paraventi, M. Wright. The Minerals, Metals and Materials Series (Springer, Cham, 2019), pp. 1949–1963
25. G. Oberson, M. Audrain, J. Collins et al., NRC perspectives on primary water stress corrosion cracking of high-chromium, nickel-based alloys. in*Proceedings of the 18th International Conference on Environmental Degradation of Materials in Nuclear Power Systems-Water Reactors*, ed. by J.H. Jackson, D. Paraventi, M. Wright. The Minerals, Metals and Materials Series (Springer, Cham,2019), pp. 55–65
26. R. Zhu, J. Wang, L. Zhang, Z. Zhang, E.H. Han, Stress corrosion cracking of 316L HAZ for 316L stainless steel/inconel 52 M dissimilar metal weld joint in simulated primary water. Corros. Sci. **112**, 373–384 (2016). https://doi.org/10.1016/j.corsci.2016.07.031
27. Z. Lu, T. Shoji, Y. Takeda, Y. Ito, A. Kai, N. Tsuchiya, Effects of loading mode and water chemistry on stress corrosion crack growth behavior of 316L HAZ and weld metal materials in high temperature pure water. Corros. Sci. **50**, 625–638 (2008). https://doi.org/10.1016/j.corsci.2007.08.021
28. J. Feng, C. Luo, J. Liu, H. Liu, Y. Zhang, X. Chen, M. Jiang, Y. Tian, Narrow gap multi-pass laser welding with 52 M/308L double wires filler addition for SA508/316L dissimilar metals. Int. J. Press. Vessels Pip. **211**, 105286 (2024). https://doi.org/10.1016/j.ijpvp.2024.105286
29. J.C. Feng, D.W. Rathod, M.J. Roy, J.A. Francis, W. Guo, N.M. Irvine, A.N. Vasileiou, Y.L. Sun, M.C. Smith, L. Li, An evaluation of multipass narrow gap laser welding as a candidate process for the manufacture of nuclear pressure vessels. Int. J. Press. Vessels Pip. **157**, 43–50 (2017). https://doi.org/10.1016/j.ijpvp.2017.08.004
30. J.C. Feng, W. Guo, N. Irvine, L. Li, Understanding and elimination of process defects in narrow gap multi-pass fiber laser welding of ferritic steel sheets of 30 mm thickness. Int. J. Adv. Manuf. Technol. **88**, 1821–1830 (2017). https://doi.org/10.1007/s00170-016-8929-1
31. W.T. Ouyang, L. Zhang, H.C. Wu, D. Wu, S.W. Zhang, X. Qin, S.L. Jiang, S.J. Li, W.W. Zhang, L.Y. Sheng, Optimized mechanical properties of the hot forged Ti-6Al-4V alloy by regulating multiscale microstructure via laser shock peening. Int. J. Mach. Tools Manuf. **201**, 104192 (2024). https://doi.org/10.1016/j.ijmachtools.2024.104192